Book strip

Science · LAUNCH · Autumn 2 · Week 1 · SCI-LAUNCH-A2-W01-L2

Use: mitosis in sequence

I am learning to use the cell-cycle sequence to find and repair errors and explain matching chromosome sets.

Success: order the stages; repair the first error; explain matching sets.

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Made by Matt

Route card

Staff: cut these apart before the lesson. Give one route card. Other routes remain available.

Supported



Supported

Supported route card with a model cell

Order interphase, mitosis and cytokinesis. Show A and B in each new cell. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Independent



Independent

Independent route card with a model cell

Order interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Mark the four phases that belong to mitosis. [4]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Aspire



Aspire

Aspire route card with a model cell

Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. [4]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Choice



Choice

Choice route card with a model cell

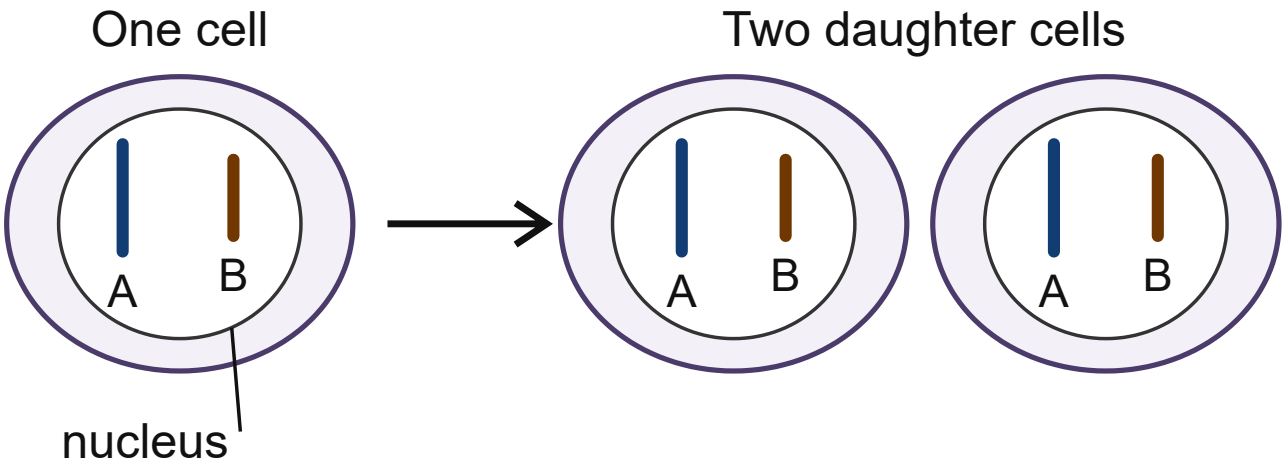
Choose the first, middle and last stage from interphase, mitosis and cytokinesis. Show both A and B in each new cell. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Made by Matt

Reference A matching chromosome sets



Compare the full A and B set in both daughter cells. Each new cell has a matching set.

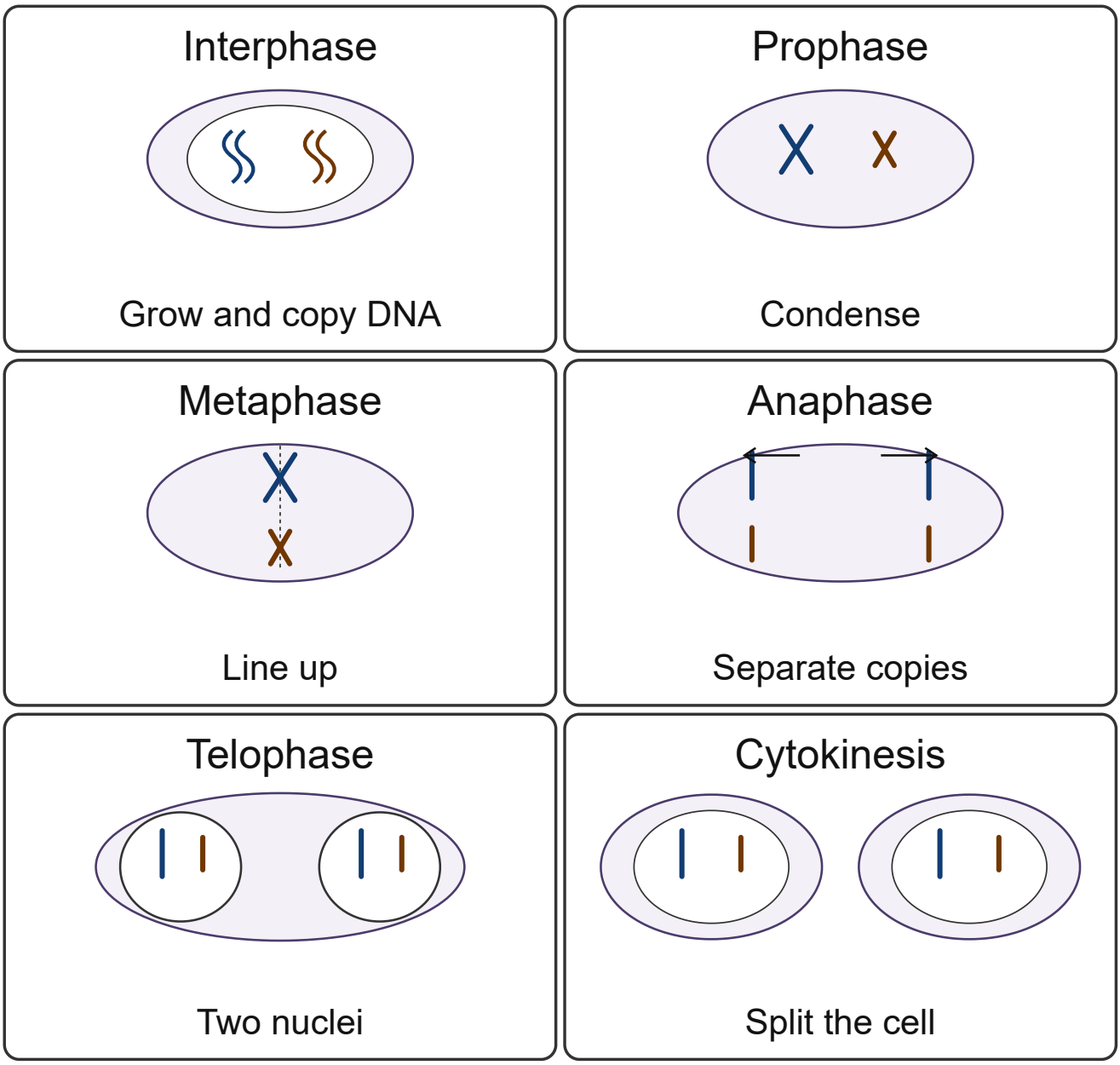
Reference only. Use labels as well as colours. The symbols are simplified chromosome types, not a full picture of real DNA.

Step	What changes
Interphase	The cell grows and copies its DNA.
Mitosis	Copied chromosomes separate into two nuclei.
Cytokinesis	The cell splits into two daughter cells.

In the number case, six chromosomes in the starting body cell means six in each daughter after ordinary mitosis and cell division. A fresh case starting with eight gives eight in each daughter.

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Reference B the six named stages



Interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Four middle phases belong to mitosis.

Reference only. Interphase comes before mitosis. Cytokinesis follows the formation of the two nuclei. The four named mitosis phases are prophase, metaphase, anaphase and telophase.

Made by Matt

Catch-up sheet absence or access only

Use only after absence or an access barrier. This replaces a starting step; it is not added missed work.

Interphase: the cell grows and copies DNA. Mitosis: copied chromosomes separate into two nuclei. Cytokinesis: the cell divides.

Use Reference A to put the three main stages in order. Check both new sets. Then join the current model or You do task 1.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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Choice task sheet

Select an answer and give a short reason. Use Reference A and B.

1. Choose the first, middle and last stage from interphase, mitosis and cytokinesis. Show both A and B in each new cell. [3] Staff I ☐ S ☐ C ☐

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2. Choose anaphase or telophase for the missing step after metaphase. Show what happens to the copies. [2] Staff I ☐ S ☐ C ☐

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3. Fault 1 skips DNA copying. Choose copy DNA first or split the cell first as the repair. Give a reason. [3] Staff I ☐ S ☐ C ☐

Made by Matt